

Mask Architecture Anomaly Segmentation for Road Scenes

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Abstract

Autonomous driving uses semantic segmentation models to understand the spatial context. Their main problem is overconfidence whenever they face Out-of-Distribution anomalies. After an evaluation of post-hoc baselines on pixel-based (ERFNet) and mask-based (EoMT) architectures, we propose, to improve the performance, the introduction of a prototype-based extension for EoMT. We extract object queries in order to compute prototypes for each specific class, making anomaly segmentation a cosine distance and a Mahalanobis distance problem. We used two different approaches: one with a global covariance matrix and the other with separate per-classes covariance matrices. The fine-tuning strategy is to use a cosine classifier to map known classes and isolate anomalies.

Code: <https://github.com/MariaVerdari/AnomalySegmentation2026.git>

1. Introduction

Semantic segmentation allows autonomous vehicles to recognize their surroundings by classifying every pixel in an image. However, current models have a critical flaw: when they encounter an unexpected obstacle or a previously unseen object (an anomaly), they tend to mistakenly classify it as a known element. This phenomenon, where the system is overconfident even when it is wrong, represents a severe danger to road safety.

Our goal is to improve current anomaly segmentation systems. We propose a new method to help models recognize when they are facing *Out-of-Distribution* (OoD) objects, thus making autonomous driving much safer and more reliable. In particular, our idea is to isolate anomalies before compressing the features into final probabilities. By defining class-specific prototypes, we use distance metrics to measure the deviation of Out-of-Distribution obstacles



Figure 1. Visual comparison between an input image from the dataset RoadAnomaly21 and the resulting heatmap after our second training.

from known representations. As illustrated in Figure 1, our extension is able to isolate OoD obstacles, assigning them high anomaly scores (highlighted in red).

2. Background

Semantic segmentation is essential for visual scene understanding. To formally explain its task: given an input image $I \in \mathbb{R}^{H \times W \times C}$, the aim is to generate a pixel-level label map $Y \in \mathcal{K}^{H \times W}$, where $\mathcal{K} = \{1, 2, \dots, K\}$ represents a set of K predefined, in-distribution (ID) classes. Semantic segmentation architectures are based on two main paradigms. ERFNet uses a pixel-based approach with convolutional layers for the classification task. In contrast, EoMT relies on a mask-based approach for the segmentation of image regions. While both methods are very effective at analyzing known objects, they share a problem: the overconfidence of the softmax function, which forces the model to give high probabilities to be in one of the known classes, even when it has never seen a specific object [7]. For a given pixel or query, the probability p_k of belonging to a specific class k is obtained by the raw logit l_k as fol-

lows:

$$p_k = \frac{\exp(l_k)}{\sum_{j=1}^{K+1} \exp(l_j)}$$

where $K + 1$ represents the optional void class. The presence of the $\exp(\cdot)$ operator amplifies even the slightest differences between the raw logits. This limitation, given by the overconfidence, underlines the necessity of searching other kinds of strategies to isolate anomalies before the compression of their final probability. The section’s aim is to give an architectural comparison between the two baselines, highlighting their limitations upon which our extension is based.

2.1. Architectural comparison between ERFNet and EoMT

The first model is **ERFNet** (Efficient Residual Factorized ConvNet), a convolutional neural network specifically designed for real-time, pixel-level semantic segmentation [11]. This architecture directly addresses the needs of intelligent vehicles, which require fast and accurate image processing despite the limited computational resources of embedded systems. Structurally, ERFNet operates on the traditional *per-pixel* paradigm: the encoder progressively downsamples the spatial dimensions of the image to gather context, while the decoder restores them through deconvolution layers to obtain an output map at the original input resolution, as shown in Figure 2. This allows to output a probability distribution among all categories for each pixel. To optimize efficiency without compromising accuracy, ERFNet introduces *residual non-bottleneck* layers that break down classic $2D\ 3 \times 3$ convolutional filters into sequential $1D$ filters of 1×3 and 3×1 dimensions. Furthermore, the encoder uses dilated convolutions to significantly expand the model’s receptive field without losing spatial resolution or increasing the number of parameters. The experiments on the Cityscapes dataset [5] demonstrate that ERFNet achieves a good trade-off between segmentation accuracy and computational efficiency. However, this *per-pixel* approach exhibits severe limitations in Out-of-Distribution (OoD) detection. Operating on a closed vocabulary of classes, the network suffers from overconfidence amplified by the softmax function.

To overcome the rigidity of pixel-level classification, we examine a different approach based on the **EoMT** (Encoder-only Mask Transformer) architecture. Instead of assigning a category label to every single pixel, EoMT adopts a mask-based paradigm, which separates the spatial division of image regions from their actual semantic classification [9]. The model relies on a *Single-Scale ViT* (Vision Transformer) that processes the input while maintaining a constant spatial resolution and a fixed patch size (usually 16×16 pixels) throughout the entire depth of the network. Unlike more complex state-of-the-art architec-

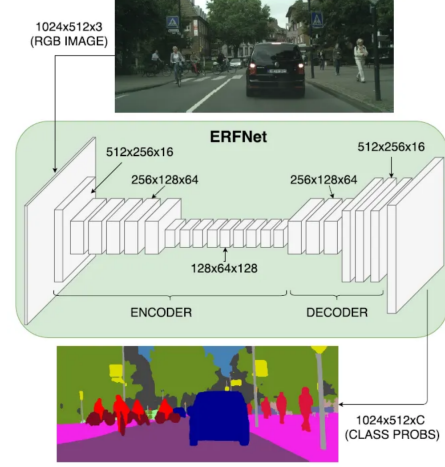


Figure 2. ERFNet architecture.

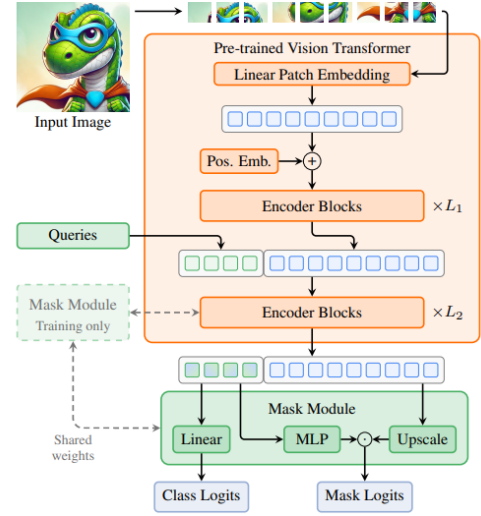


Figure 3. EoMT architecture.

tures like *Mask2Former* [3, 4], which combine the ViT with heavy modules such as multi-scale adapters and pixel decoders to refine edges, EoMT demonstrates that large-scale pre-training provides sufficiently dense information to make these extra components unnecessary. The model identifies objects directly through a fixed set of 100 learnable *Object Queries*, which are concatenated to the image *Patch Tokens* within the final blocks of the encoder (Figure 3). As a result, the interaction between the queries and the image happens entirely in the same space through *Self-Attention* mechanisms, eliminating the need for separate *Cross-Attention* modules or masked attention strategies during inference. By leveraging initialization from *Vision Foundation Models* (VFM) like DINOv2 [10], EoMT achieves good generalization and robustness against out-of-distribution scenarios.

3. Post-hoc anomaly segmentation baseline methods

3.1. Methodology

To establish the initial baselines for our work, we apply post-hoc anomaly detection methods directly to the two models’ predictions. Without altering the internal architectures or the pre-trained weights of the networks, this strategy relies strictly on modifying the evaluation and inference logic to capture raw output scores. Since ERFNet and EoMT operate on fundamentally different architectural paradigms—the former being a pixel-based model and the latter a mask-based framework—their outputs must be processed through distinct post-hoc inference strategies:

- In the case of ERFNet, the pixel-structured output enables a direct post-hoc scoring process: the network’s logits l_k are extracted at the pixel level to compute multiple baseline anomaly scores.
- In the case of EoMT, the mask-structured output requires a dedicated semantic inference process: for each query, the network returns a mask logit and a class logit. We apply a sigmoid function to the mask logit to obtain a “mask probability” $M_n(x) \in [0, 1]$ at pixel x , which quantifies the degree to which the pixel belongs to that segment. In parallel, a softmax function is applied to the class logit to obtain the “class probability” $P_{n,k}$ for query n and class k , excluding the extra *void* category to discard empty masks. These outputs are then combined to map the query-level predictions back to the pixel space for each class:

$$L_k(x) = \sum_{n=1}^N P_{n,k} M_n(x) \quad (1)$$

Finally, the resulting tensor is interpolated to achieve the target spatial resolution of 1024×1024 , matching the input dimensions.

Before detailing the mathematical formulations used to convert these outputs into final anomaly scores, we define the metrics selected to assess their anomaly segmentation performance. To rigorously measure the effectiveness of the various methodologies against dataset-specific ground-truth masks, we do not rely on standard accuracy. Indeed, on highly imbalanced datasets (where anomalous pixels represent only a small fraction of the entire road scene), traditional accuracy can be misleadingly high due to the overwhelming dominance of the in-distribution background. Therefore, we use dedicated pixel-level metrics that are robust to class imbalance:

- The first is **AuPRC** (Area Under the Precision-Recall Curve), which evaluates the balance between precision and recall at various sensitivity thresholds. The higher the AuPRC is, the better trade-off between precision and recall is achieved.

- The second is the **FPR95** (False Positive Rate at 95% True Positive Rate) metric, which quantifies the percentage of false anomalies detected when the threshold is set to guarantee the detection of 95% of actual anomalies. Minimizing FPR95 is a strict requirement for autonomous driving safety, as it prevents unjustified braking caused by normal road patches being mistakenly classified as obstacles.

Complementary to the anomaly segmentation metrics, we also evaluate the standard in-distribution performance of the models. This allows us to ensure that the network maintains its primary semantic segmentation quality, which we measure via the mean Intersection over Union (**mIoU**) on the Cityscapes dataset. For a specific class c , the Intersection over Union (IoU_c) calculates the overlap between the ground-truth region G_c and the predicted mask P_c .

The overall segmentation accuracy is then summarized by the mean IoU (mIoU), computed as the average IoU over all C classes.

Having defined the evaluation criteria, we now introduce the specific post-hoc scoring functions. To quantitatively calculate an anomaly score $S(x)$ for each pixel, we implement and compare several distinct methodologies:

- The Maximum Softmax Probability (MSP) evaluates the confidence of the most likely class, where low values indicate model uncertainty, signaling a potential anomaly. To ensure that a higher anomaly score for a pixel corresponds to a greater likelihood of an anomaly, the score is defined by subtracting the maximum probability from one. However, this method remains highly susceptible to overconfidence due to the inherent calibration issues of the softmax function.
- The Max Logit method directly analyzes the logits l_k before applying the softmax function, thereby better preserving the raw information about the network’s confusion.
- The Max Entropy method quantifies informational disorder by considering the entire probability distribution using the Shannon entropy formula; high entropy indicates that the network is distributing its uncertainty across multiple known classes, which is a typical behavior when facing an OoD input.
- To exploit the properties of EoMT’s mask-based architecture, we implement the Rejected by All (RbA) scoring function [12]. This method computes the anomaly score by summing the logits (calculated as in Eq. 1) of all K known classes after mapping them through a hyperbolic tangent ($\tanh$) activation function. The underlying intuition relies on the network’s behavior when encountering out-of-distribution anomalies: if a pixel is rejected by all known semantic categories, all its corresponding logits drop close to zero. Consequently, the sum of their

hyperbolic tangents vanishes, resulting in an RbA score near zero. Conversely, for normal in-distribution pixels, at least one known class will exhibit a highly positive logit, driving its tanh value toward one and producing a strongly negative (and therefore lower) anomaly score.

- Finally, for the MSP baseline we integrate Temperature Scaling into the inference pipeline: by dividing the logits by a temperature parameter, we mitigate the softmax overconfidence. The anomaly score is computed as:

$$S_{MSP;T}(x) = 1 - \max_{k \in \{1, \dots, K\}} \frac{\exp(l_k/T)}{\sum_{j=1}^K \exp(l_j/T)}.$$

Since a higher temperature softens the otherwise over-confident class distribution, it can alter the relative ordering of pixel scores and therefore affect the metrics. We vary T over the values $\{0.5, 0.75, 1.0, 1.1, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0\}$, where $T = 1.0$ recovers the standard MSP. For each T we sum the quantity (AuPRC – FPR95) over all the evaluation datasets and pick the temperature that maximises this aggregate objective.

3.2. Experiments and results for post-hoc methods

The evaluation of the previously pre-trained models relies on different datasets which provide binary pixel-level ground-truth anomaly masks. In particular, we evaluated on the original *RoadAnomaly* dataset, alongside *RoadAnomaly21* and *RoadObstacle21* (both from the *SegmentMeIfYouCan* benchmark [2]), which feature anomalous objects in diverse driving scenarios. We use also, from *Fishyscapes* [1], *FS Lost&Found* with real-world objects physically put on the road and *FS Static* with synthetically blended out-of-context anomalies into real validation images.

Following the evaluation protocol of the ERFNet baselines, we adopt a fixed-resolution resize rather than sliding-window inference. We set the input resolution to 1024×1024 to match EoMT’s training resolution (the ERFNet baselines instead use 512×1024 , their native resolution). Sliding-window inference could give different results since it avoids resizing the images but it is a computationally heavier alternative.

The experimental results presented in the tables highlight several key findings regarding the effectiveness of the applied post-hoc scoring methods for anomaly segmentation:

- **Architectural Superiority of EoMT:** Across all evaluated datasets (Tables 1 and 2), the mask-based EoMT consistently and significantly outperforms the pixel-based ERFNet. This substantial gap underscores the advantage of modern mask-based architectures in handling out-of-distribution features compared to traditional pixel-based models.

- **Post-hoc Method Trends for ERFNet:** When analyzing ERFNet, the Max Logit metric proves to be the most effective strategy. As shown in Tables 1 and 2, by operating directly on the raw logits rather than the normalized probabilities, it avoids the overconfidence distortions typically introduced by the softmax function.
- **Post-hoc Method Trends for EoMT:** Conversely, for the EoMT model, Max Entropy emerges as the most robust scoring mechanism, consistently providing the highest AuPRC while maintaining competitive false positive rates (see Tables 1 and 2). Furthermore, while the RbA approach achieves reasonable AuPRC scores, it suffers from severe degradation in FPR on specific datasets.
- **Impact of Temperature Scaling:** The temperature scaling experiments demonstrate that adjusting the temperature parameter t does not alter the primary semantic segmentation performance for either architecture. As mathematically expected, both ERFNet (Tables 3 and 4) and EoMT (Tables 5 and 6) successfully maintain their constant mIoU scores; since temperature scaling strictly acts on the logits’ magnitude before the softmax function, it preserves their relative ordering and leaves the final *argmax* class predictions entirely unchanged. Shifting the focus to anomaly segmentation, the two models exhibit distinct behaviors when applying different temperatures. For ERFNet, scaling the temperature slightly above 1.0 (with $t = 1.1$ identified as the optimal value) yields marginal but consistent improvements, while lowering the temperature actively degrades performance. Conversely, the mask-based EoMT proves to be remarkably robust to minor temperature variations, showing only fractional fluctuations in AuPRC and FPR. Notably, its optimum is reached only at the largest tested temperature ($t = 5.0$), although the overall gain remains negligible.

Table 1. Performance comparison of post-hoc methods using ERFNet and EoMT

Model	Method	SMIYC RA-21		SMIYC RO-21	
		AuPRC	FPR	AuPRC	FPR
ERFNET	MSP	29.10	62.55	2.71	65.22
	MaxLogit	38.32	59.34	4.63	48.44
	MaxEnt	30.97	62.66	3.04	65.91
EoMT	MSP	76.25	34.44	89.29	0.37
	MaxLogit	75.75	34.56	89.29	0.37
	MaxEnt	79.24	34.33	89.39	0.40
	RbA	72.06	78.89	85.38	99.90

4. Prototype-based extension for anomaly segmentation

To advance the anomaly segmentation capabilities of EoMT, we propose a *prototype-based* extension that moves

Table 2. Performance comparison of post-hoc methods using ERFNet and EoMT

Model	Method	FS L&F		FS Static		Road Anomaly	
		AuPRC	FPR	AuPRC	FPR	AuPRC	FPR
ERFNET	MSP	1.75	50.59	7.47	41.84	12.42	82.58
	MaxLogit	3.30	45.49	9.50	40.30	15.58	73.25
	MaxEnt	2.58	50.16	8.84	41.55	12.67	82.75
EoMT	MSP	26.83	12.65	54.84	36.12	77.44	20.12
	MaxLogit	26.90	12.52	55.04	35.87	76.89	19.88
	MaxEnt	30.27	12.53	53.72	36.06	79.73	19.64
	RbA	27.78	7.46	57.42	80.48	76.59	19.90

Table 3. ERFNet Temperature Scaling

Method	mIoU	SMIYC RA-21		SMIYC RO-21	
		AuP.	FPR	AuP.	FPR
MSP	72.17	29.10	62.55	2.71	65.22
$MSP(t = 0.5)$	72.17	27.06	62.73	2.42	63.23
$MSP(t = 0.75)$	72.17	28.16	62.49	2.57	64.13
$MSP(t = 1.1)$	72.17	29.40	62.65	2.76	65.87
MSP (best $t = 1.1$)	72.17	29.40	62.65	2.76	65.87

Table 4. ERFNet Temperature Scaling

Method	mIoU	FS L&F		FS Static		Road Anomaly	
		AuP.	FPR	AuP.	FPR	AuP.	FPR
MSP	72.17	1.75	50.59	7.47	41.84	12.42	82.58
$MSP(t = 0.5)$	72.17	1.28	66.74	6.60	43.48	12.19	82.02
$MSP(t = 0.75)$	72.17	1.49	51.76	6.99	42.50	12.32	82.31
$MSP(t = 1.1)$	72.17	1.86	50.18	7.69	41.62	12.46	82.73
MSP (best $t = 1.1$)	72.17	1.86	50.18	7.69	41.62	12.46	82.73

Table 5. EoMT: Temperature Scaling

Method	mIoU	SMIYC RA-21		SMIYC RO-21	
		AuP.	FPR	AuP.	FPR
MSP	77.10	76.25	34.44	89.29	0.37
$MSP(t = 0.5)$	77.10	76.13	34.44	89.28	0.37
$MSP(t = 0.75)$	77.10	76.20	34.44	89.29	0.37
$MSP(t = 1.1)$	77.10	76.27	34.44	89.29	0.37
MSP (best $t = 5.0$)	77.10	76.63	34.44	89.30	0.37

Table 6. EoMT: Temperature Scaling

Method	mIoU	FS L&F		FS Static		Road Anomaly	
		AuP.	FPR	AuP.	FPR	AuP.	FPR
MSP	77.10	26.83	12.65	54.84	36.12	77.44	20.12
$MSP(t = 0.5)$	77.10	26.81	12.66	54.81	36.12	77.28	20.18
$MSP(t = 0.75)$	77.10	26.83	12.65	54.83	36.12	77.37	20.14
$MSP(t = 1.1)$	77.10	26.84	12.65	54.85	36.12	77.46	20.12
MSP (best $t = 5.0$)	77.10	26.91	12.64	54.86	36.12	77.86	20.08

beyond the standard post-hoc analysis. Instead of deriving anomaly scores from the output probabilities, our method operates directly in the high-dimensional feature space of the model’s object queries.

The underlying premise is that, after training, the object queries assigned to in-distribution classes concentrate

around a set of per-class centroids, which we refer to as *prototypes*. Conversely, queries that capture out-of-distribution (OoD) objects deviate considerably from all known class distributions (see Figure 4). This formulation draws conceptual inspiration from classical statistical distance metrics, such as the Mahalanobis distance, which evaluate outliers by accounting for the underlying distribution’s shape and dispersion. By framing the anomaly detection task around the statistical representativeness of object queries within the learned feature manifold, an anomalous query therefore behaves as an outlier in feature space, and its distance from the class prototypes provides a measurable anomaly signal.

To realise this framework, we modify the forward pass in order to access the object queries computed by EoMT. After the final transformer block, we apply the backbone’s output normalisation and extract the first 100 tokens from the token sequence (the green blocks just before the mask module in Figure 3), recovering the updated queries as D -dimensional feature vectors. These vectors serve as the input to the distance-based scoring described in the following sections.

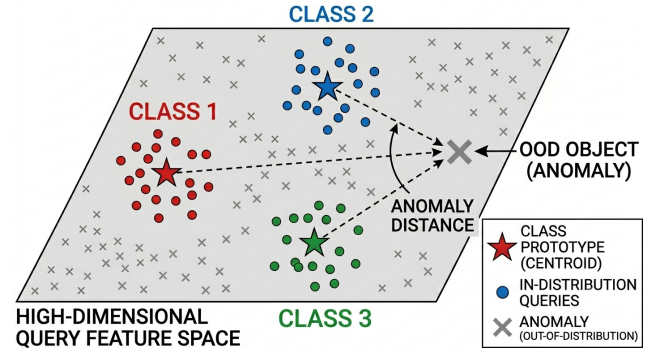


Figure 4. Conceptual overview of the query space. While in-distribution object queries cluster around their respective per-class prototypes, an OOD anomaly deviates significantly from all known class distributions. The resulting distance from these centroids provides a measurable anomaly signal.

4.1. Prototype extraction

The initial phase of this extension involves computing a prototype for every known category. We perform a forward pass over all the available Cityscapes images (train, validation and test splits, $\sim 5,000$ images in total) using the frozen, pre-trained EoMT model. For every processed image, we capture the 100 query embeddings alongside their corresponding class logits. To ensure that the prototypes reflect only clean and reliable class representations, we apply a strict confidence threshold, discarding any query whose maximum prediction probability is below 70%, as well as any query assigned to the “no-object” class.

The remaining valid queries are grouped by their predicted class label, and for each category c the class prototype μ_c is defined as the mean of its query embeddings. In addition, we estimate a single global covariance matrix Σ over the pooled set of all valid queries, centered on their global mean. We initially considered per-class covariance matrices Σ_c , but a shared covariance proved preferable: it is estimated from a much larger sample, treats all classes uniformly, and requires a single matrix inversion at test time. To guarantee a well-conditioned inversion, we regularize the estimate by adding a small constant to the diagonal ($\Sigma \leftarrow \Sigma + 0.01 \cdot I$). These statistics are computed once and stored, acting as the reference distribution for the anomaly scores described next.

4.2. Distance-based anomaly scores

At inference time on the anomaly datasets the model shifts from predicting classification probabilities to computing distance-based anomaly scores. Each of the 100 object queries q receives a score measuring how far it lies from all known class prototypes. We experiment with two distances.

The first is the Mahalanobis distance, computed against every prototype μ_c and reduced with a minimum over the classes:

$$D_M(q) = \min_c \sqrt{(q - \mu_c)^T \Sigma^+ (q - \mu_c)},$$

where Σ^+ is the pseudo-inverse of the global covariance matrix. Unlike the Euclidean distance, which assumes a spherical distribution, the Mahalanobis distance accounts for the variance and correlation of the feature dimensions, so that directions with high natural variance do not artificially inflate the score.

The second is the cosine distance, which discards vector magnitudes and only compares directions:

$$D_{cos}(q) = \min_c (1 - \cos(q, \mu_c)). \quad (2)$$

In an earlier design, each query was compared only against the prototype of its predicted class. We found it preferable to take the minimum over all classes instead: an anomalous query may be confidently assigned to a wrong known class, so anchoring the score to the unreliable prediction weakens the signal, whereas an anomaly is more robustly characterized as being far from every known prototype.

4.3. Query-to-pixel broadcasting

Once each query is mapped to a distance score, the resulting vector of 100 per-query scores must be projected back onto the spatial grid of the image. We leverage the predicted masks: applying a sigmoid to the raw mask logits yields, for each query, a per-pixel assignment probability.

Each pixel is then assigned the anomaly score of its winning query, i.e., the query with the highest mask probability at that coordinate. The resulting anomaly map is upsampled with bilinear interpolation to the 1024×1024 working resolution, and used both to generate qualitative heatmaps and to compute the evaluation metrics.

We also explored a soft alternative, where each mask probability map is multiplied by its query score and the per-pixel maximum over the weighted maps is taken. The hard winning-query assignment proved preferable: with the soft variant, a pixel can inherit a high score from a query whose mask only weakly overlaps it, diluting the spatial precision of the anomaly map.

4.4. Prototype-guided fine-tuning

The distance-based scores described above rely on the geometry of the query space, which the original training never explicitly optimized. Inspired by PAnS [6], we therefore fine-tune the model so that this geometry is shaped directly by the prototypes. The key modification is architectural: the linear classification head is replaced by a *cosine classifier*, whose weight rows act as learnable prototypes. It uses cosine similarity between the input features and the class weights as class scores. While PAnS computes prototype similarities at the pixel level on dense feature maps, we transfer the same mechanism to the query level of EoMT. The class logits therefore become:

$$\ell_c(q) = \tau \cdot \cos(q, w_c) = \tau \frac{q \cdot w_c}{\|q\| \|w_c\|},$$

with temperature $\tau = 10$, computed for the 19 known classes plus the “no-object” category. The cosine scores are in the range $[-1, 1]$ due to the normalization effect on the denominator, but the temperature maps them in the range $[-\tau, \tau]$. Each row w_c is initialized with the corresponding precomputed prototype μ_c .

The training objective follows the original EoMT framework. We maintain the standard bipartite Hungarian matching and the localization objectives (Mask and Dice losses) to refine object boundaries. The only difference lies in the classification head. Instead of standard linear logits, the classification cross-entropy now operates on the temperature-scaled cosine similarities:

$$\mathcal{L}_{cls}(q, y) = -\log \frac{\exp(\tau \cos(q, w_y))}{\sum_c \exp(\tau \cos(q, w_c))}. \quad (3)$$

The standard cross-entropy turns into the prototypical loss of PAnS through the minimization of this objective. It pulls each matched query q towards the prototype of its ground-truth class w_y while pushing it away from all other prototypes.

The fine-tuning starts from the pre-trained EoMT weights. The ViT backbone, including the blocks that process the queries, is kept entirely frozen [8]; only the query

embeddings, the mask head, the upscale modules and the classification head are updated. We compare two variants: in Training 1, the cosine head is trained jointly, letting the prototypes move away from the initial class means (*learned prototypes*); in Training 2, the weight matrix of the cosine head is frozen after being initialized with the prototypes μ_c , so that the loss can only move the queries towards the fixed class means (*fixed prototypes*). After fine-tuning, anomalies are scored with the PAnS rule:

$$S = 1 - \max_c \frac{1}{2}(\cos(q, w_c) + 1)$$

with the head weights w_c in place of the precomputed prototypes μ_c . This formulation is simply a monotone transformation with respect to Eq. 2.

4.5. Experiments and results

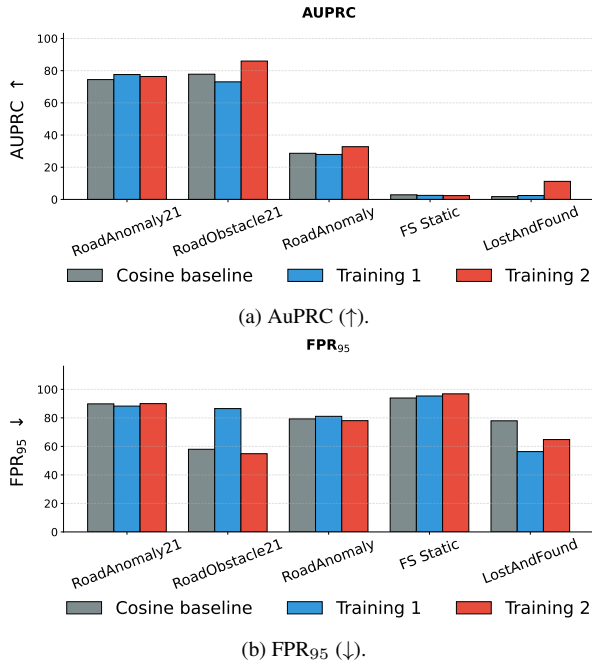


Figure 5. Effect of prototype-guided fine-tuning across the five datasets.

The model is fine-tuned for 15 epochs on the Cityscapes training split (2,975 images), with AdamW optimizer, an initial learning rate of 10^{-4} with a short two-stage warmup followed by polynomial decay, weight decay 0.05, and an effective batch size of 16 (batch size 2 with 8 gradient-accumulation steps), using 16-bit mixed precision. The validation mIoU on Cityscapes is monitored every 5 epochs as a sanity check that fine-tuning does not degrade the closed-set segmentation quality. We evaluate our proposed extension on the same suite of driving anomaly datasets introduced in Section 3.2.

The joint analysis of performance metrics (Table 7), AuPRC and FPR95 trends shown in Figure 5 and qualitative visualizations (Figure 6) outlines critical insights regarding the effectiveness of the various prototype-based strategies and the two subsequent fine-tuning steps applied to the EoMT model.

- Among purely post-hoc methods based on distances from prototypes, Cosine similarity and Mahalanobis distance with a shared covariance matrix prove to be markedly superior to the classic per-class Mahalanobis formulation on high-contrast datasets (RoadAnomaly21 and RoadObstacle21). Although Cosine similarity provides a competitive anchor in terms of AUPRC, the Shared Σ variant exhibits remarkable robustness in containing false positives across extended OoD scenarios, preserving the best FPR₉₅ values on both RoadAnomaly and LostAndFound. The FS Static dataset is confirmed as the most challenging environment for the entirety of methods, where only the Mahalanobis with per-class covariance matrix approach manages to extract a marginal advantage.
- The introduction of dedicated training phases yields an increase in out-of-distribution (OoD) discrimination capacity compared to traditional statistical-geometric metrics. Training 2 emerges as the most robust configuration for maximizing AUPRC. This behavior peaks on the RoadObstacle21, RoadAnomaly and LostAndFound datasets. Conversely, Training 1 demonstrates strong efficacy in macroscopic anomaly scenarios such as RoadAnomaly21, scoring the best combination in that column.
- A key insight emerges when cross-referencing the quantitative metrics with the qualitative heatmaps in Figure 6. Notably, for several input images, traditional post-hoc baselines already deliver highly robust anomaly detection and complete structural coverage without requiring any training. On these specific samples, the fine-tuning process fails to provide an advantage and instead degrades mask quality by triggering a severe “hollowing-out” effect. This is particularly evident in larger anomalies (such as the giraffe’s neck) where the trained models erroneously absorb the object’s interior into the in-distribution feature space, leaving prominent false-negative holes within an anomaly mask that untrained methods capture uniformly.

5. Conclusion

In this work we studied anomaly segmentation on road scenes by comparing two complementary segmentation paradigms, the pixel-based ERFNet and the mask-based EoMT and we proposed a prototype-based extension of EoMT that operates directly in the space of its queries. Under standard post-hoc scoring, the mask-based EoMT consistently outperformed ERFNet across all five dataset. Building on these baselines, we reframed anomaly detec-

Table 7. Comprehensive evaluation of anomaly segmentation performance across different prototype-based post-hoc methods and fine-tuning experiments. The best results for each dataset are highlighted in **bold** (AUPRC $\uparrow$, FPR₉₅ $\downarrow$).

Dataset	Per-class Mahal.		Shared Σ Mahal.		Cosine		Training 1		Training 2	
	AUPRC	FPR ₉₅	AUPRC	FPR ₉₅	AUPRC	FPR ₉₅	AUPRC	FPR ₉₅	AUPRC	FPR ₉₅
RoadAnomaly21	57.80	89.78	70.99	89.80	74.46	89.82	77.59	88.27	76.39	90.00
RoadObstacle21	76.67	60.29	76.37	61.82	77.80	57.98	73.06	86.52	85.97	54.88
RoadAnomaly	27.42	70.56	26.76	70.32	28.67	79.29	27.95	81.09	32.77	78.05
FS Static	4.10	93.03	3.60	98.97	2.87	93.94	2.52	95.35	2.37	96.86
LostAndFound	2.05	56.88	1.59	54.11	1.75	77.94	2.43	56.34	11.22	64.79

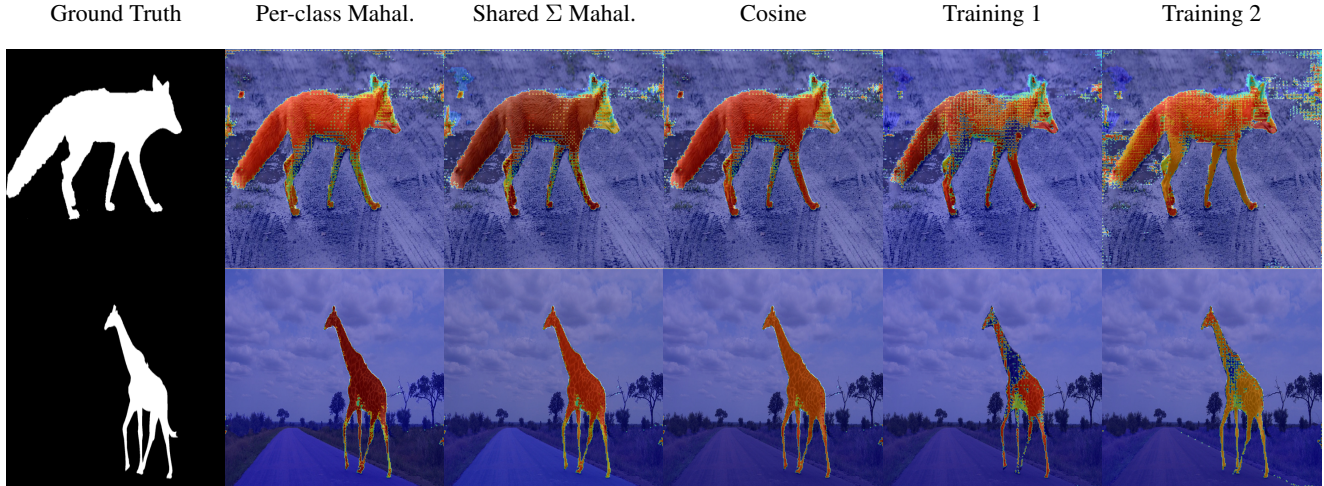


Figure 6. Qualitative comparison of ground truth and anomaly heatmaps across different experimental conditions for representative samples from RoadAnomaly (top row) and RoadAnomaly21 (bottom row).

tion as a measure of the distance of the queries with respect to a set of class prototypes computed on Cityscapes. Among the purely post-hoc distances, the cosine similarity and the shared-covariance Mahalanobis distance were the most reliable. Prototype-guided fine-tuning, which reshapes the query geometry through a cosine classifier in the spirit of PANs, further improved discrimination on several datasets: the fixed-prototype variant (Training 2) achieved the strongest AuPRC overall, most notably a large gain on LostAndFound.

Although no single method dominates across all datasets, several promising directions emerge as potential open questions. An idea could be re-training the model without freezing the backbone, but this solution is computationally expensive. Another solution could be to explicitly teach the model how to recognize novel elements integrating an Outlier Exposure and anomaly injection pipeline into the framework. By pasting diverse object from external datasets directly onto the Cityscapes training images, the network could hypothetically be exposed to proxy anomalies during the optimization phase. This pipeline might then be governed by a dedicated loss function incorporating a

specific regularization term; such term would aim to actively push the embeddings of the injected anomalies away from the calculated in-distribution class prototypes, thereby potentially widening the distance gap. Additionally, exploring a systematic hyperparameter tuning of the scaling parameter τ remains an open possibility to further investigate its impact on the calibration margin, offering a speculative direction toward self-calibrated mask-based architectures in safety-critical driving scenarios.

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